Lecture 9: Blackboard Notes (Animal Collective Behavior)

A suggested sequence of boards (two of them are short live derivations). Each block lists what to write; sketches to draw are noted in italics. The minimal models (Vicsek, Toner-Tu) are recalled from Lecture 7 and used here as tools to read experiments.

Opening: what this lecture is about (motivation before Board 1)

Thousands of birds, fish, or bacteria move as one with no leader and only local information. Lectures 7 and 8 gave us the *models* (Vicsek, Toner-Tu). **This lecture is the bridge to the living system: how to measure, quantify, and infer collective behaviour in real animal groups – and how those measurements pushed back on the theory.**

Three jobs run through the boards:

- **measure / quantify**: turn "it looks coordinated" into numbers – the order parameter Φ and the correlation function $C(r)$ (Board 1)
- **infer**: read the interaction rules straight out of trajectory data, using maximum entropy (Boards 3-4)
- **test**: confront the models with experiment – and several times the data won, forcing the theory to change (Boards 5-7)

What the measurements revealed (surprises vs equilibrium).

- **True long-range order in two dimensions**, which the Mermin-Wagner theorem forbids in equilibrium – possible only because the group is driven, far from equilibrium.
- **Scale-free correlations**: the correlation length grows with the group, so flocks behave as if poised at a **critical point** (apparently tuned by evolution, not by an experimenter).
- Information travels as an **undamped wave**, faster than any single animal's reaction time – not the slow diffusion a naive model predicts (Board 6 vs Board 7).
- Interactions are **topological** (a fixed number of neighbours), not metric (a fixed distance) – a result that came from data, not from theory.

Board 1: Quantifying collective motion

- **polarization order parameter** ($\hat{\mathbf{v}}_i = \mathbf{v}_i/|\mathbf{v}_i|$):

$$\Phi = \frac{1}{N}\left|\sum_{i=1}^{N} \hat{\mathbf{v}}_i\right| \in [0,1] \qquad (0 = \text{disorder},\ 1 = \text{aligned})$$

- **velocity-fluctuation correlation** ($\delta\mathbf{v}_i = \mathbf{v}_i - \langle \mathbf{v}\rangle$):

$$C(r) = \frac{\langle \delta\mathbf{v}_i \cdot \delta\mathbf{v}_j\rangle_{|r_{ij}|=r}}{\langle |\delta\mathbf{v}|^2\rangle}$$

- correlation length ξ = first zero of $C(r)$
- flocks are **scale-free**: $C(r) = f(r/L)$, so $\xi \sim L$ (correlations span the whole group $\Rightarrow$ near a critical point)

Board 2: Vicsek model (recall from Lecture 7)

$$\theta_i(t+1) = \arg\Big(\sum_{j\in\mathcal{N}_i} e^{i\theta_j}\Big) + \eta\,\xi_i, \qquad \mathbf{r}_i(t+1) = \mathbf{r}_i + v_0(\cos\theta_i, \sin\theta_i)$$

- control parameters: noise η, density ρ
- order-disorder transition; in fact **first order** (traveling bands at onset)

Board 3: Maximum entropy – principle and constraints

- **the problem**: from tracking data we can reliably measure only **low-order statistics** – each bird's mean heading, and how correlated *pairs* of birds are. What is the simplest model consistent with them, without inventing extra rules?
- **maximum-entropy principle** (Jaynes): choose the distribution $P(\{\mathbf{s}\})$ that maximizes the entropy

$$S = -\sum_{\{\mathbf{s}\}} P(\{\mathbf{s}\})\,\ln P(\{\mathbf{s}\})$$

subject to the measured constraints

- entropy = spread / uncertainty ⇒ maximizing it = the **least-biased** ("most honest") model: it assumes *nothing beyond the data*. Any lower-entropy model has smuggled in structure the data do not justify
- *same principle as equilibrium statistical mechanics*: maximize S at fixed mean **energy** → Boltzmann distribution; here we fix measured **correlations** instead
- constraints ($\mathbf{s}_i = \hat{\mathbf{v}}_i$ = unit heading of bird i):
    1. normalization $\sum P = 1$
    2. single-particle means $\langle \mathbf{s}_i\rangle = \mathbf{m}_i$ (measured)
    3. pair correlations $\langle \mathbf{s}_i \cdot \mathbf{s}_j\rangle = C_{ij}$ (measured)
- **why stop at pairs?** triplets and higher need far more data; the real test is whether pairs *alone* already predict the rest (next board)

Board 4: Derivation – maximum entropy gives a Boltzmann form

- Lagrange function:

$$\mathcal{L} = S + \lambda_0\Big(1 - \sum P\Big) + \sum_i \lambda_i \cdot (\mathbf{m}_i - \langle \mathbf{s}_i\rangle) + \sum_{i<j} \Lambda_{ij}(C_{ij} - \langle \mathbf{s}_i \cdot \mathbf{s}_j\rangle)$$

- set $\partial\mathcal{L}/\partial P = 0 \;\Rightarrow\; \ln P = \text{const} + \sum_i \lambda_i \cdot \mathbf{s}_i + \sum_{i<j} \Lambda_{ij}\,\mathbf{s}_i \cdot \mathbf{s}_j$
- exponentiate ⇒ a Boltzmann / Heisenberg-spin model:

$$P(\{\mathbf{s}_i\}) = \frac{1}{Z}\exp\Big(\sum_{i<j} J_{ij}\,\mathbf{s}_i \cdot \mathbf{s}_j + \sum_i \mathbf{h}_i \cdot \mathbf{s}_i\Big)$$

- **punchline**: "what we measured (1- and 2-point statistics)" ⇒ an Ising/Heisenberg model; fit $J_{ij}, \mathbf{h}_i$ to data
- **the result (Bialek et al.)**: a model fit to *pair* correlations alone correctly predicts the **higher-order** statistics (triplets, the full distribution of the order parameter) ⇒ pairwise interactions are enough to explain the flock
- the inferred parameters place real flocks **near a critical point** (maximal responsiveness)

Board 5: Topological vs metric interactions

- **metric**: all neighbors within a fixed radius $R \Rightarrow$ count varies with density
- **topological**: the k nearest neighbors $\Rightarrow$ fixed count, density-independent
- starlings (Ballerini 2008): $k \approx 6$ to 7
- Voronoi cells define neighbors; mean number of Voronoi neighbors = 6 in 2D (Euler) $\Rightarrow$ why $k \approx 6$

Sketch: a focal bird in a dense vs a sparse patch – fixed-radius circle (count changes) vs k-nearest (count fixed).

Board 6: Derivation – why standard models propagate information diffusively

- ordered phase: all headings nearly aligned. Write $\theta_i = \theta_0 + \phi_i$, with ϕ_i a small **turning fluctuation**
- the Vicsek rule is "**align to the neighbour average**": each bird relaxes its heading toward the mean of its neighbours

$$\dot{\phi}_i \;\propto\; \langle\phi\rangle_{\text{neighbours}} - \phi_i$$

- but (neighbour average − own value) is exactly what the **Laplacian** measures $\Rightarrow$ in the continuum:

$$\partial_t \phi = D\,\nabla^2\phi \qquad (D \sim \text{alignment strength})$$

this is the **diffusion (heat) equation** for the turning field - test one Fourier mode $\phi \sim e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}$: use $\partial_t \to -i\omega,\ \nabla^2 \to -q^2$, so $-i\omega = -Dq^2$

$$\Rightarrow \qquad \omega = -iDq^2$$

- ω is **purely imaginary** $\Rightarrow$ no oscillation, no travelling wave: $e^{-i\omega t} = e^{-Dq^2 t}$, every mode just **decays in place**
- so a disturbance **spreads, it does not travel**: $x \sim \sqrt{Dt}$ (like heat in a metal bar), not $x \sim vt$
- consequence: crossing a flock of size L takes time $\sim L^2/D$ (grows with L^2!) – far too slow; a turn washes out before it reaches the far side and **fragments** a large flock $\Rightarrow$ need a better model (Board 7)

Board 7: Information propagation – sound modes and behavioral inertia

- real flocks: turning info travels **linearly and almost undamped**
- add **behavioral inertia** (resistance to changing heading) $\Rightarrow$ a second time derivative, i.e. a wave equation:

$$\partial_t^2\phi = c_s^2\,\nabla^2\phi \qquad \Rightarrow \qquad \omega = c_s q$$

- speed $c_s > v_0$ (non-local transfer); dynamic exponent $z \approx 1$ (vs $z \approx 2$ for diffusion in Board 6)
- this is the Toner-Tu "sound" of a flock; confirmed in starling turns (Attanasi 2014: $c_s \approx 20$ to 40 m/s, $v_0 \approx 10$ to 15 m/s)

Board 8: Universality across species

- birds, fish, locusts, ants, bacteria
- **shared**: order-disorder transition, density threshold, scale-free correlations, fast information transfer
- **differs**: the interaction channel (vision, hydrodynamics, chemotaxis, contact)